Five structural tables related to A_6

Private L054 recovery component

Theorem 1. *Every displayed structural description in the five current companion tables S_5, A_6, M_{10}, $P\Sigma L_2(9) \cong S_6$, and $P\Gamma L_2(9)$ holds for the actual subgroup class specified by the retained numerical row map. The tables have respectively $19, 22, 25, 56, 65$ rows, hence 187 descriptions in total. An extension label asserts the indicated decomposition, without claiming that its factor names determine a unique isomorphism type.*

Source and acquired interfaces. The tables are attributed to Thomas Connor and Dimitri Leemans. All eight native source pages were directly read. Current PDF bytes, not an assumed historical edition identity, define these assertions. Their complete numerical classifications and source row maps are acquired from the accepted robust-weight, M_{10}, and final-six components. The filenames `psigmal29` and `pgammal29` are locators; the source titles and exact native row syntax are recorded separately.

Products associate from the left, following the acquired naming article. A dot means non-splitting, a colon a split extension, and a cross a direct product. For example the $P\Gamma L_2(9)$ rows 6, 9 and 13 are $(((3 : S_3) \cdot 2) \cdot 2) : 2$, $((3 : S_3) \cdot 2) \cdot 2$, and $(D_{10} \cdot 2) \times 2$. Row 17 is $Q_8 : 2^2$, not $Q_8 : 22$ with cyclic complement of order 22. The native elementary abelian powers $2^2, 2^3, 3^2$ are explicit syntax nodes. The dihedral subscript denotes the total group order.

Embedded representative reconstruction. The M_{10} saved array and the $P\Gamma L_2(9)$ completed state contain literal embedded generators. Some earlier S_5, A_6, S_6 arrays instead print `SymmetricGroup` or `AlternatingGroup` on an explicit point set. We do not assume that a display name preserves the producer's hidden embedding. Those expressions supply candidate reconstructions: the full symmetric or alternating group on exactly that point set, fixing its complement. All other function expressions are rejected.

Here is the independent check that pays this boundary. Close each candidate subgroup literally and check membership in the actual ambient permutation group. Enumerate its conjugacy orbit under the ambient generators, as sets of literal permutations. These orbits are pairwise disjoint, have the acquired class lengths, and their number equals the acquired complete number of subgroup classes. Consequently their union is the complete actual subgroup set. For each retained representative, select its maximal proper members in this finite set by descending order and subset tests. Count them by ambient class. If the downward count is a and the two ambient class lengths are c_H, c_K, the upward count is ac_H/c_K by counting incidence pairs. The checker verifies integrality and both counts against every source pair, including nonedges. This proves the actual source join without an assumption about the old printed embeddings. All 8,831 ordered pairs pass; no new ambient subgroup search is used.

Reusable structural certificates. Forty concrete certified models are acquired from the preceding PSL_2, PGL_2, semilinear, field-extension and Mathieu/alternating atom packets. In particular M_{10} and M_9 retain their exact point-stabilizer interfaces in the standard M_{11} action; they are not identified by orders 720 and 72 alone. Two elementary additional models are S_4, S_6: the literal groups on four and six letters have orders $4!, 6!$, so equal the full symmetric groups on those sets. These are not classifications inferred from the orders of arbitrary groups.

Ten further models carry actual decompositions. Every factor is an acquired or earlier verified model. A source occurrence is identified with its model by a literal bijection fixing identity and preserving all right-generator edges. The acquired MC.3 argument proves that such a bijection is a homomorphism, by induction on words. The same argument transports every factor certificate. Identical printed labels can index several different models; a successful actual bijection is always required.

The split tests check a normal subgroup N, a complement K, their verified factor types, $N \cap K = 1$ and $|N||K| = |H|$. A noncommuting pair of factor generators verifies nontrivial action. Direct nodes additionally require all cross-generator pairs to commute. These conditions prove the claimed products. A dotted prime-index node is checked by listing every outside element and requiring that its pth power is nonidentity: a complement at index p exists exactly when an outside element has order p. The only new dotted model has index two and eight outside elements. Other dotted descriptions use acquired model transports. No failed complement search is a nonsplitting proof.

Finite evidence and validation. The literal reader using the acquired criteria checks all 187 row joins, ten new models, 206 transports and 31,324 right-generator edges. Candidate generation uses only the retained actual groups. Its changed, bounded pass returned in 3.45 seconds. An earlier unused recursive candidate had a trivial split action; the unchanged literal reader rejected it. That failed check and its exact code are preserved. Candidate selection was strengthened to require nontrivial action, with no relaxation of the final criterion.

Damaged certificates test identity, bijective nonhomomorphisms, missing or duplicate rows and elements, altered actual generators/maps, powers, dots, association, factors and cyclic dependencies. Additional controls reject duplicate reconstructed classes, a changed source weight, a missing ambient generator list and unsafe named expressions. Independent direct product and prime-index examples distinguish the positive and negative cases. Ordinary and optimized Python must agree. No discovery routine is rerun by the frozen checker.

Scope. The proof acquires the prior complete numerical classifications and named model definitions explicitly. Its new result is the 187 exact current structural descriptions and their embedded source joins. It neither asserts unique extension names nor completes all structural atlas rows. No original-source, question, formalization or measurement count changes here. The author prepared the finite application under root's registry reuse instruction; the standard structural and transport arguments are acquired, not claimed as new mathematics.